

mission during PCB design. This involves precise positioning and routing of PCBs, adhering to specific design specifications. Every aspect of PCB design, from the components used to the assembly process and the arrangement and interconnection of components, plays a crucial role. A solid understanding of signal integrity principles is essential for designing high-speed electronics. Following best practices ensures the optimal performance of high-speed electronics.

Basics of signal integrity

Various factors can cause the quality of a signal on a circuit board to deteriorate. Industry experts and studies have identified four primary signal integrity issues

1. Electromagnetic interference (EMI). If high-frequency signals are not appropriately routed on a circuit board, they can emit EMI, similar to a poorly tuned radio station. The configuration and length of traces and via stubs can act as antennas and cause problems. Additionally, the signal return path can be a source of EMI; it is best to have it on an adjacent reference plane. If the signal return path is blocked, the signal will emit more noise as it tries to find a path back to its source.

2. Unintentional electromagnetic coupling (crosstalk). High-speed signals routed too closely can unintentionally interfere with one another, causing crosstalk. This interference can cause the affected signal to mimic the interfering signal, failing to perform its intended function. Crosstalk is an issue with adjacent signals running side-by-side and routed in parallel on adjacent circuit board layers. This type of interference, called broadside coupling, is why most circuit board designs use alternating routing directions horizontally and vertically across neighbouring layers.

3. Simultaneous switching noise (ground bounce). Due to numerous circuit elements transitioning between high and low states, the voltage level may not quickly return to the intended ground potential when transitioning to a low state. If the low state's voltage rebounds excessively high, it could erroneously register the low state as high. Multiple occurrences can lead to false or double switching, causing disruptions in the circuit's operation.

4. Impedance mismatch. Variations in the consistency of sensitive high-speed transmission lines can lead to signal reflections that disrupt the signal's integrity. Traces routed without proper attention to their impedance value are susceptible to changes in those values in different board areas based on various conditions. Adequate routing for impedance-controlled sensitive traces requires specific layer arrangements, trace widths, and clearances.

Impact of poor signal integrity

Interference can significantly impact the functionality of electronic equipment. For example, a poorly designed PCB layout causing signal integrity issues can result in a computer crash, a phone call drop, or a data transfer failure. Poor signal integrity often leads to subtle issues, such as performance hiccups, data loss, or complete device failure. These problems become more pronounced as signal speed increases, making electronics more vulnerable to interference like signal reflections from impedance mismatches, ground bounce, and crosstalk. Without deliberate layout considerations to mitigate these issues, signal degradation may persist until the circuit no longer functions as intended.

Best PCB layout practices

To ensure optimal signal integrity in PCB design, consider the follow-

ing guidelines:

- Ensure signal path traces are short and direct
- Route sensitive signals on internal layers, ideally positioned between reference planes
- Separate high-speed signals from other traces whenever feasible. Use ground or power planes as shields to isolate sensitive signals from other traces, reducing crosstalk and improving signal integrity
- Length match groups of nets by starting with the longest connection and adjusting others with tuning turns to match it
- Avoid routing sensitive signals through noisy areas, such as analogue or power supply sections of the board
- Allow sufficient space for specific routing configurations like daisy chains, if necessary
- Minimise the use of vias to reduce their length and inductance, which could affect signal integrity

In addition to routing, designing the board's power delivery network (PDN) is essential for maintaining power integrity and ensuring good signal integrity. Avoid routing high-speed transmission lines through blocked areas on the reference plane, as this can increase EMI by causing signals to meander around, seeking a clear return path to their source.

Signal integrity resides in the white space of circuit schematics, where engineers might not always pay close attention. As electronic systems demand higher data rates, it is no longer enough to make the right net connections or select the right components. Designers must pay greater attention to layout, which significantly impacts signal integrity. Poorly designed layouts increase the cost of fixing issues and extend the time to market, often requiring a complete re-spin of the board design. **EFY**